

mapping brain activity by recording magnetic fields produced by electrical currents occurring naturally in the brain. Functional magnetic resonance imaging (fMRI) measures brain activity by detecting changes associated with blood flow. This technique is based on the fact that cerebral blood flow and neuronal activation are coupled, such that when an area of the brain is in use, the blood flow to that region increases.

MEG and fMRI are used as non-invasive BCIs. In an experiment, fMRI based BCI was used by two respondents to play Pong (an electronic game that simulates table tennis by using in-game paddles to hit the ball back and forth) in real time by altering their brain blood flow through biofeedback techniques. fMRI has also been used to control robot arms with a seven-second delay between thought and movement.

Brain-to-brain communication system

In a brain-to-brain communication system two participants, Brain 1 and Brain 2, located about 2km apart communicate using EEG based non-invasive BCIs to play a video game. Brain 1 can see the game but has no controller. Brain 2 has a controller but cannot see the game. When wanting to make a move, Brain 1 is required to simply think about moving his hand and the brain signals produced are picked up by a cap of electrodes worn on his head and transmitted to a TMS device placed on the head of Brain 2. The TMS beams the signal into Brain 2. The stimulation automatically makes Brain 2's hand to move on the touch pad to manipulate the controller as thought by Brain 1. The communication takes a split second and the success rate varies from 25 to 83%, depending on the pair of people participating.

BCI APPLICATIONS

Patients with disorders of consciousness

For patients suffering with disorders of consciousness, the end-of-life decisions are made by family members considering the patient's best interest. For the patients who are able to

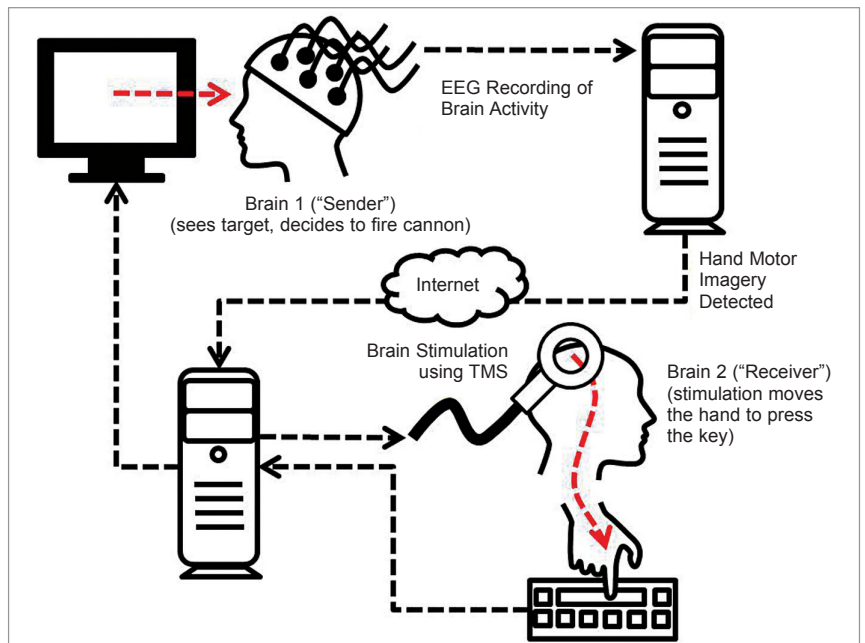


Fig. 7: Brain-to-Brain communication system (Source: www.independent.co.uk)

perform basic cognitive tasks, the BCI technology could make available communication tools to enable them make these decisions as per their desire.

However, such patients may not be able to use BCIs that are based on vision, auditory or vibrotactile stimuli, as there are chances of the patients fading in and out of consciousness. In some cases, it may become difficult to determine when they would respond, or the response may last for a very little time only.

Motor imagery BCIs for neurorehabilitation

Motor imagery is a cognitive process in which a person imagines performing a movement without performing it, or even tensing his/her muscles. In motor imagery based BCIs the user generates the induced activity from motor cortex by imagining motor movements without any limb movement or external stimulus. Motor imagery BCIs serve as an interface to communicate the user's intention without limb movement.

The most representative application is for neurorehabilitation. The University of Melbourne is developing a system to enable patients with paralysis to control external devices such as robotic limbs, computers, and exoskeletons by translating their brain activity.

Functional brain mapping

Functional brain mapping procedure is often required during neurosurgery for patients with tumors or epilepsy, for which electrodes are placed on the brain to precisely identify the locations of structures and functional areas. New ways are being explored for using BCI technology to improve the mapping using non-invasive means.

Neurogaming

Neurogaming involves the use of BCIs so that the users can interact with the game without the use of a traditional controller. BCIs detect and infer the gamer's intent. Intensive training is required by the users to acquire the acceptable ability to master the control system.

BCIs for armed forces

Direct brain-to-brain communication offers the most transformative application for the armed forces for operational use. The US Army has initiated a research project to explore the possibility of creating a technology that will enable soldiers to convey messages by thought, for use in covert operations. The project involves recording one's thoughts, transmitting the recording to a particular individual who decodes it as a voice or text.